



Healthcare Duplicate MRN Continuous Improvement Analysis

Using Python, SQL, and Power BI to Identify Opportunities
for Reducing Duplicate MRNs

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Healthcare Analytics Portfolio

July 2026



What problem are we trying to solve?

Business Problem

Duplicate medical record numbers (MRNs) create operational inefficiencies, increase duplicate investigation workload, and can contribute to patient safety risks if patient information is fragmented across multiple records.

Project Goal

Identify the highest-impact process improvement opportunities for reducing duplicate MRNs through analysis of duplicate investigation data.



How was the solution developed?

Workflow

- Business Problem
- Domain Knowledge
- Python Simulation
- SQL Analysis
- Power BI Dashboard
- Recommendations

Technology Stack

- Python (data simulation)
- SQL (business analysis)
- Power BI (visualization)

Why synthetic data?

Synthetic data was modeled after realistic healthcare registration workflows to demonstrate an end-to-end analytics solution while protecting patient privacy.



How was the solution built?



Python Data Generation

- 1,000 synthetic duplicate investigations
- Weighted probability simulation
- Source-specific business rules



Healthcare Workflow Rules

- Mutually exclusive contributing factors
- Realistic healthcare registration workflows
- Categorized contributing factors into five primary categories



SQL Analysis

- Business question-driven SQL queries
- Aggregated trends using GROUP BY, COUNT, and ORDER BY
- Summarized trends by source, department, contributing factor, and month



Power BI

- Two-page executive dashboard
- Interactive Opportunity Explorer
- Continuous improvement-focused reporting



What did the analysis reveal?

Total Duplicate MRNs

1,000

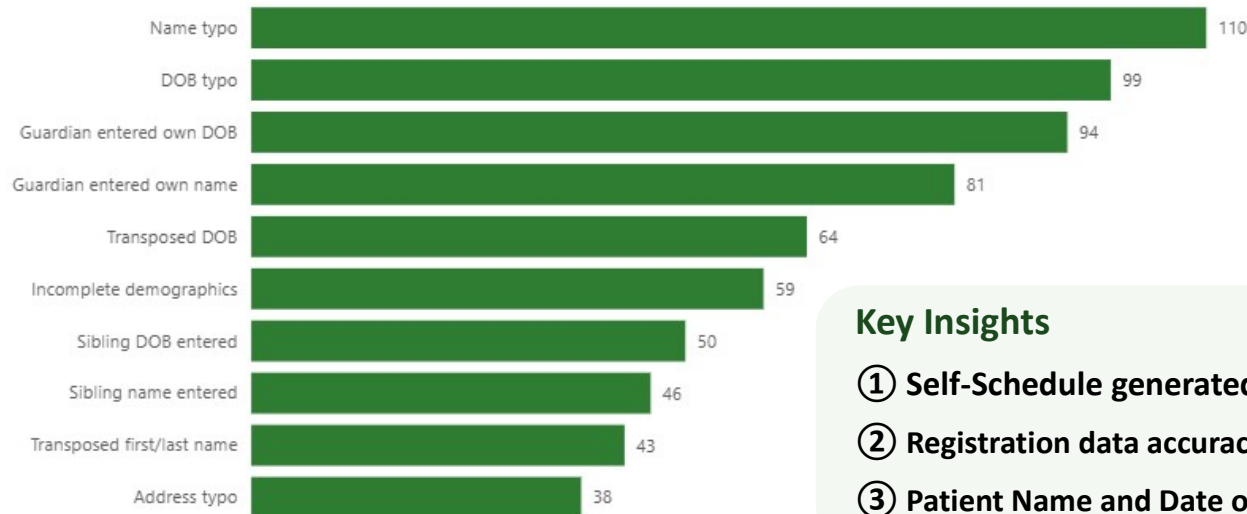
Primary Duplicate Source

Urgent Care (34.5%)

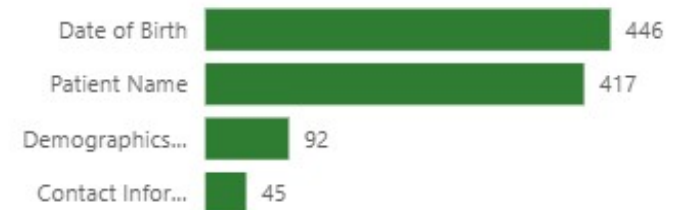
Top Contributing Factor

Address typo

Top Improvement Opportunities



Contributing Factor Categories



Key Insights

- ① Self-Schedule generated the highest duplicate volume (34.5%).
- ② Registration data accuracy represents the largest improvement opportunity.
- ③ Patient Name and Date of Birth accounted for most duplicate investigations.

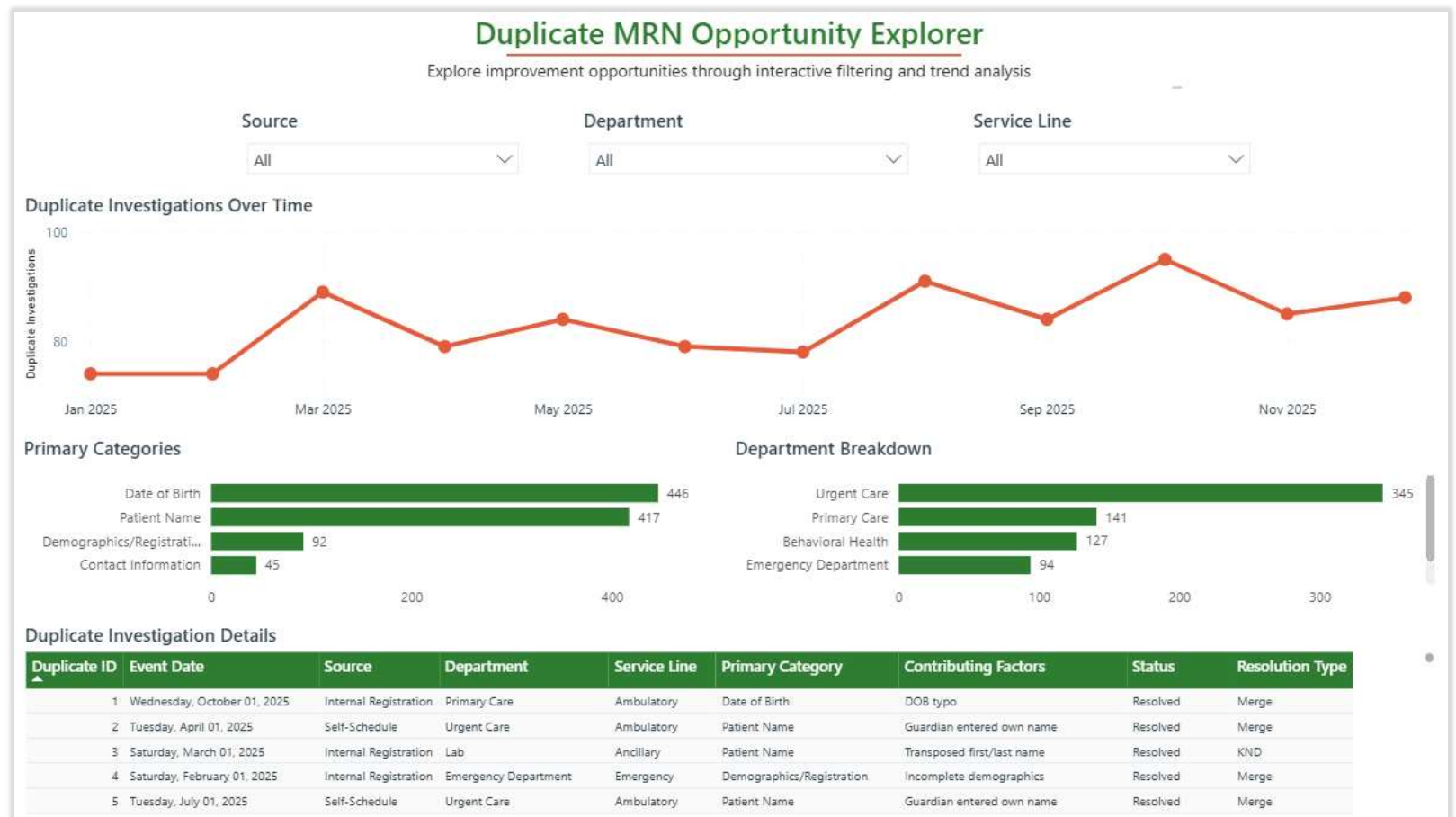
🔍 How can stakeholders explore improvement opportunities?

Opportunity Explorer

Enables department leaders to investigate duplicate MRN patterns through interactive filtering and drill-down analysis.

Capabilities

- ✓ Compare duplicate trends by source
- ✓ Identify high-volume departments
- ✓ Explore service-line trends
- ✓ Monitor trends over time
- ✓ Review investigations details





Recommended Process Improvements

Reinforce Reliable Search Method

Improve staff consistency when searching for existing patient records before creating new MRNs.

Reduces preventable duplicate record creation.

Reinforce STAR Safety Tool

Practice STAR (Stop, Think, Act, Review).

Encourage staff to pause and verify patient identifiers before completing registration.

Reduces data entry errors.

Target High Duplicate Creators

Assign Duplicate MRNs & Overlays training.

Require staff with higher duplicate creation rates to complete refresher education.

Supports targeted performance improvement.

Monitor Department Performance

Provide leaders with real-time monitoring tools.

Use dashboards and quarterly rounding to review duplicate trends and coach teams.

Sustains continuous improvement.

Thank you / Questions

Thank you for your time and feedback.

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